

Adaptive Learning Systems: Transforming Pedagogical Approaches

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Abstract

Adaptive learning systems, which tailor content, pacing, and feedback to individual students' unique needs, offer personalized learning experiences. This customization is enabled by sophisticated learning analytics that continuously collect and analyze students' performance and interactions with learning materials to optimize the learning process. As a result, students benefit from self-paced and self-directed learning opportunities, fostering greater engagement and autonomy. This article examines the architecture and functioning of adaptive learning systems with a particular focus on leveraging Artificial Intelligence (AI) and machine learning. It explores the practical applications of these systems in language classrooms and beyond, addressing both the opportunities and challenges they present. The article concludes with suggestions for future directions in the field.

Keywords

Adaptive learning systems, computer-assisted language learning, intelligent systems, personalized learning, Artificial Intelligence, Large Language Models, ChatGPT

Introduction

One of the critical challenges in language classrooms is addressing learner variability, as learners come with diverse knowledge levels, learning goals, and styles, making their learning needs and preferences unique (Slavuj et al., 2017). The prevalent ‘one-size-fits-all’ approach, in which a single instructional method or material is applied to all, can become less effective, often irrelevant, and unengaging for learners. Promoting *personalized learning* is thus essential, as research evidence has found considerable learning benefits for groups of learners who were taught in a personalized manner according to their needs (VanLehn, 2011). However, it is not feasible for an individual human instructor to tailor a curriculum to many individual learners.

Adaptivity in instructional systems has thus emerged as a practical solution, catering to each learner’s unique needs and characteristics. An adaptive learning system can be broadly defined as a system that adjusts instruction based on identified characteristics of learners to improve the efficiency and efficacy of learning (Oxman et al., 2014).

Elements of Adaptive Learning Systems

The definition of an adaptive learning system suggests the technological basis of adaptive systems that typically includes three elements: learner model, domain model, and adaptation model (Brusilovsky, 2012; Knutov et al., 2009). The **learner model** is the source of adaptation, representing the learner’s characteristics to which the system adapts, such as learning styles, performance, and proficiency levels. The **domain model** is the target of adaptation, defining what is to be learned through a space consisting of a set of knowledge elements. The adaptation model connects these two models and determines how the learning experience should be tailored. Table 1 provides a detailed overview of these three models.

Table 1*The Architecture of an Adaptive System*

Component	Function
Learner Model (<i>Adapt to what?</i>)	Keeps the learner model up-to-date by analyzing and interpreting user behaviours and inferred learning outcomes (i.e., learner modeling)
Domain Model (<i>Adapt what?</i>)	Establishes clear relationships between learning materials and domain items (i.e., knowledge indexing)
Adaptation Model (<i>How to adapt?</i>)	Implements the adaptation by combining the information from the learner and domain models.

Methods of Adaptation

A common method of adaptation is that the system *selects* and *orders* the *content* to be presented to the learner based on the learner's needs and characteristics identified. Additionally, problem-solving support is particularly advantageous in the context of language learning, as any in-time help the system offers to the learner. It provides scaffolding and hints, immediate feedback on correct and incorrect answers, and misconception/error identification on learner solutions (Slavuj et al., 2017). Applications that function in such a way include popular language-learning apps.

Language Learning App

Language learning is well-suited for adaptive learning approaches due to the diverse needs and varying proficiency levels of learners from different backgrounds. With the popularity of informal language learning, there is a great demand for personalized approaches, both in traditional classrooms and self-directed environments. Due to space constraints, we have included only Duolingo as a representative example of language learning apps, though other apps like Babbel, Rosetta Stone, and Memrise also employ similar adaptive learning features.

Duolingo, one of the most popular language learning apps, has over 500 million active users worldwide, offering courses in over 30 languages (Duolingo, 2024). It features a structured course with a sequence of lessons and exercises created by human experts and AI. Lessons are presented as a linear ‘path’ divided into smaller ‘units’ with specific learning objectives. Learners take lessons in the order presented on the path, but what each learner sees in a lesson is unique since Duolingo uses AI models to tailor each lesson to the learner’s needs and knowledge level (Pajak & Bicknell, 2022).

Upon signing up, learners take a placement test, allowing them to start at an appropriate level and skip basic units. Lessons adapt in real-time, adjusting difficulty based on performance. More correct answers lead to more challenging exercises, while incorrect answers prompt additional exercises focusing on those mistakes (Vazquez et al., 2020). Another adaptive feature is spaced repetition, where the algorithm schedules reviews based on the learner’s retention needs, ensuring personalized practice (Jiang et al., 2024).

Artificial Intelligence and Machine Learning

Compared to well-defined learning domains like mathematics, however, language learning, being both skill and knowledge-based and following a nonlinear process, is more complex to construct and implement into adaptive systems (Garrett, 2009; Godwin-Jones, 2017). Due to this nature, language instructional systems often employ Machine Learning (ML) and Artificial intelligence (AI) techniques and methods. ML techniques use the observations of learner’s actions and behaviour to develop the training set for making future predictions, providing adaptivity, as seen in Duolingo (Troussas et al., 2019). AI techniques, particularly Natural Language Processing (NLP), are used to process a solution for a learner’s free language output using such corpus analysis and grammatical error simulations as in Intelligent Tutoring Systems

(ITSs). ITSs have been built using these technologies to achieve the effect of one-on-one tutoring that a human teacher can provide. ITSs, using their “intelligence,” identify the abilities and weaknesses of each student to base their problem-solving support and sequencing learning items and tasks (Hasanov et al., 2019; Slavuj et al., 2017). ITSs have sometimes been reported to produce learning effects comparable to human tutoring (VanLehn, 2011). However, the level of engagement and feedback provided by these systems remains unsatisfactory (Kuo et al., 2023) due to challenges such as difficulties in evaluating productive skills like writing and speaking, as well as in providing nuanced feedback beyond simple correct-incorrect statements (Slavuj et al., 2015).

Recent advances in Large Language Models (LLMs), like OpenAI’s ChatGPT, Microsoft’s CoPilot, and Google’s Gemini, suggest new ways to enhance adaptive learning mechanisms and students’ learning experiences. For instance, while multiple-choice items were previously favored in evaluating students’ success with learning items due to their straightforward dichotomous (right or wrong) nature, evaluating learners’ free responses to open-ended questions was challenging due to technological limitations (Kwon, 2023). Rather than providing instant and personalized feedback that could enhance students’ learning, the adaptivity of these earlier systems was often merely a list of desired learning materials for an individual in their subsequent learning. LLMs, pre-trained on vast amounts of data, are highly sophisticated and make it possible to understand and predict language patterns, generating human-like text. These models now allow for automated and more detailed examination of students’ extended responses and direct identification of misconceptions. More adaptive systems are starting to incorporate LLMs, with the GPTs’ API accessible for constructing chatbots within the adaptive platform. These chatbots interact with students to produce meaning-focused

dialogues (Meniado, 2023) and qualitative information that helps more nuanced diagnosis of students' weaknesses, which allows for scaffolding, personalized feedback for remediation, and the generation of customized assessment items (Kuo et al., 2023).

The new Duolingo Max, powered by GPT-4, offers enhanced personalized learning (Duolingo Team, 2023). The 'Explain My Answer' chat feature allows learners to ask more about the lesson and receive explanations and examples to clarify why an answer was right or wrong. The 'Roleplay' feature allows learners to practice real-world conversations in various scenarios, with world characters, such as ordering coffee at a cafe in Paris or asking a friend to go for a hike. After the interaction, learners receive feedback on the accuracy and complexity of their responses, as well as tips for future conversations. Currently, these features are available for English learners of Spanish, French, and Finnish, with plans to expand to more languages.

Future considerations

Most adaptive systems focus on areas like vocabulary, grammar, and reading; however, systems targeting other skills (i.e., listening, speaking, and writing) are relatively scarce (Kaur et al., 2023; Slavuj et al., 2017). These systems typically adapt to learners' performance by measuring the number of correct and incorrect answers to questions (Kaur et al., 2023). Essentially, they rely heavily on quantitative learning data, yet a more comprehensive understanding of student learning may require qualitative data, especially in dealing with language production (Ayedoun et al., 2020). That said, these systems should extend beyond performance metrics to include other individual learner profiles, such as learning styles and preferences, enhancing the learning experience. Typically, these characteristics have been identified through questionnaires (Truong, 2016); however, learning systems should aim to incorporate AI-based automatic detection methods to minimize biases in self-reported data (Kaur et al., 2023).

With the highly sophisticated LLMs and their associated chatbots now available, language learning platforms can leverage these models to dynamically generate personalized learning experiences that adjust to a learner's progress, preferences, and needs in real time. Among all, the ability of LLMs to address extended responses is particularly beneficial, providing feedback tailored to each student's writing rather than being generally applicable. This feedback adeptly diagnoses strengths and weaknesses concerning both global aspects (e.g., content, organization, and style) and local issues (e.g., language and grammar errors), offering specific remedies. However, despite these advantages, the feedback can often be overwhelming and may not feel as tailored to individual learners as human feedback (Na et al., 2025), indicating that further improvements are necessary.

To this end, systematic studies on student interactions with AI tools are crucial. For instance, Na et al. (2025) found that when university ESL students used ChatGPT for writing revisions, general prompts (e.g., 'Can you give me feedback on this essay?') often led to lengthy, less actionable feedback, and learners struggled to locate errors due to the lack of visual features (e.g., Grammarly's flagging). In contrast, more specific prompts (e.g., 'What do you think of the flow of the second paragraph?') resulted in more targeted, helpful feedback. However, many prompts still led to ChatGPT generating fully revised texts, even when not requested, encouraging copying and pasting behaviors from the learners.

More research is needed to optimise AI tools by enhancing features that promote engagement and effective learning behaviours, rather than simply providing easy shortcuts for students. Developers should consider integrating LLMs into their platforms to maximize learning benefits by customising the model to provide just the right amount of support, and the type and

amount of feedback learners wish to receive. However, successful implementation of LLMs in adaptive systems relies on the accuracy and precision of these models.

Pedagogical Considerations

Adaptive technology should be seen as a complement, not a replacement, for classroom instruction. The success of teaching with an adaptive platform depends on deliberate human planning, monitoring, and intervention, making the instructor's role crucial (Cavanagh et al., 2020). Instructors should guide students' self-directed learning by regularly checking in on their progress using learning analytics (e.g., login frequency, learning rates/speed, and progress) and providing supplementary materials and instructor-led feedback to address the student's weaknesses identified by the system.

Chatbots powered by LLMs offer numerous advantages for language teaching and learning. A simple prompt to ChatGPT can generate customized lesson plans, materials, and exercises tailored to students' varying needs and interests (Koraishi, 2023). Teachers can also encourage students to engage in real-time language practice with ChatGPT, receiving instant, personalized feedback on grammar and word choice (Amin, 2023). Furthermore, by specifying learning objectives, teachers can use ChatGPT to generate various assessment items, such as multiple-choice and open-ended questions, at different proficiency levels (Ockey, 2024). The benefits of formative assessment are enhanced with timely and frequent feedback from ChatGPT on student responses, fostering increased autonomy and motivation (Warschauer & Grimes, 2008).

However, AI use in student work poses challenges to academic integrity. Perkins (2023) notes that these tools can blur the lines between legitimate academic assistance and misconduct, such as plagiarism, due to the difficulty in detecting misuse. Nevertheless, if students clearly

state any AI use in their submissions, it may not necessarily be considered plagiarism (Perkins, 2023). Meanwhile, institutions should develop clear policies to address academic integrity concerns, and teachers should train students on ethical AI usage.

Conclusion

This article explored how adaptive language systems, particularly those enhanced by AI and LLMs like ChatGPT, hold transformative potential for personalized language education by providing customized learning experiences through adaptive content, assessments, and feedback to meet individual learners' needs. As these technologies continue to evolve, ongoing research and development will be crucial to maximize their potential for language learning while addressing ethical considerations, particularly around academic integrity. The role of educators is vital in ensuring these tools are integrated responsibly and effectively within educational settings. Collaborative efforts among researchers, educators, policymakers, and developers will be essential to navigate the challenges and fully realize the opportunities presented by these emerging tools.

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